INCH-POUND
MIL-W-16878/12D(NAVY)
11 September 1992
SUPERSEDING
MIL-W-16878/12C(NAVY)
10 August 1981

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, FLUORINATED ETHYLENE PROPYLENE (FEP) INSULATED, 200°C, 1000 VOLTS

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-16878.

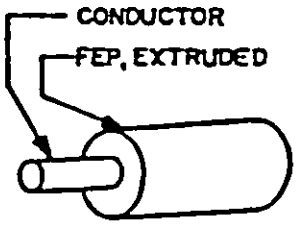


FIGURE 1. Construction.

AMSC N/A FSC 6145

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TABLE I. Construction details.

(See footnotes at end of table.)

PIN 3/	Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)	
			Material 2/	Coating		Min	Max
M16878/12-BAA*	32	1 X 32	Copper	Silver	0.0080	0.033	0.041
M16878/12-CAA*	32	1 X 32	C.C. steel	Silver	.0080	.033	.041
M16878/12-BAB*	32	7 X 40	Copper	Silver	.010	.035	.043
M16878/12-BBA*	30	1 X 30	Copper	Silver	.0100	.036	.044
M16878/12-CBA*	30	1 X 30	C.C. steel	Silver	.0100	.036	.044
M16878/12-BBB*	30	7 X 38	Copper	Silver	.012	.038	.046
M16878/12-BCA*	28	1 X 28	Copper	Silver	.0126	.039	.047
M16878/12-CCA*	28	1 X 28	C.C. steel	Silver	.0126	.039	.047
M16878/12-BCB*	28	7 X 36	Copper	Silver	.015	.041	.049
M16878/12-BDA*	26	1 X 26	Copper	Silver	.0159	.042	.050
M16878/12-CDA*	26	1 X 26	C.C. steel	Silver	.0159	.042	.050
M16878/12-BDB*	26	7 X 34	Copper	Silver	.019	.045	.053
M16878/12-BDE*	26	19 X 38	Copper	Silver	.020	.045	.053
M16878/12-BEA*	24	1 X 24	Copper	Silver	.0201	.046	.054
M16878/12-CEA*	1/ 24	1 X 24	C.C. steel	Silver	.0201	.046	.054
M16878/12-BEB*	24	7 X 32	Copper	Silver	.024	.050	.058
M16878/12-BEE*	24	19 X 36	Copper	Silver	.025	.050	.058
M16878/12-BFA*	22	1 X 22	Copper	Silver	.0254	.051	.060
M16878/12-CFA*	1/ 22	1 X 22	C.C. steel	Silver	.0254	.051	.060
M16878/12-BFB*	22	7 X 30	Copper	Silver	.030	.056	.064
M16878/12-BFE*	22	19 X 34	Copper	Silver	.032	.056	.064
M16878/12-BGA*	20	1 X 20	Copper	Silver	.0320	.058	.066
M16878/12-BGB*	20	7 X 28	Copper	Silver	.038	.064	.072
M16878/12-BGE*	20	19 X 32	Copper	Silver	.040	.064	.072
M16878/12-BHA*	18	1 X 18	Copper	Silver	.0403	.066	.077
M16878/12-BHB*	18	7 X 26	Copper	Silver	.048	.074	.084
M16878/12-BHE*	18	19 X 30	Copper	Silver	.050	.074	.084
M16878/12-BJA*	16	1 X 16	Copper	Silver	.0508	.077	.089
M16878/12-BJE*	16	19 X 29	Copper	Silver	.057	.083	.095
M16878/12-BJF*	16	26 X 30	Copper	Silver	.060	.086	.098
M16878/12-BKA*	14	1 X 14	Copper	Silver	.0641	.090	.106
M16878/12-BKE*	14	19 X 27	Copper	Silver	.072	.098	.114
M16878/12-BKH*	14	41 X 30	Copper	Silver	.076	.102	.118
M16878/12-BLE*	12	19 X 25	Copper	Silver	.091	.117	.133
M16878/12-BLJ*	12	65 X 30	Copper	Silver	.093	.119	.135
M16878/12-BMG*	10	37 X 26	Copper	Silver	.111	.137	.153
M16878/12-BMK*	10	105 X 30	Copper	Silver	.120	.146	.162
M16878/12-BNL*	8	133 X 29	Copper	Silver	.169	.199	.219
M16878/12-BPL*	6	133 X 27	Copper	Silver	.213	.286	.301
M16878/12-BRL*	4	133 X 25	Copper	Silver	.269	.349	.369

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PIN 3/	Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)	
			Material 2/	Coating		Min	Max
M16878/12-BSP*	2	665 X 30	Copper	Silver	0.315	0.395	0.415
M16878/12-BTR*	1	817 X 30	Copper	Silver	.385	.475	.495
M16878/12-BUS*	0	1045 X 30	Copper	Silver	.410	.504	.524
M16878/12-BWT*	00	1330 X 30	Copper	Silver	.455	.553	.577
M16878/12-BYY*	000	1672 X 30	Copper	Silver	.535	.633	.657
M16878/12-BZW*	0000	2109 X 30	Copper	Silver	.595	.703	.727

1/ Inactive for new design.
2/ C.C. stands for copper-clad.
3/ PIN stands for part or identifying number.

ADDITIONAL REQUIREMENTS:

Visual and mechanical examination: Required.
Spark test: 5.0 kV.
Impulse dielectric test: 8.0 kV.
Dielectric withstanding voltage: 3.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C.
K = 50,000.
D = Maximum average diameter of finished wire.
d = Conductor diameter.

Cold bend: Condition 4 hours at minus 65 ± 1°C (see table II).

TABLE II. Cold bend mandrel sizes.

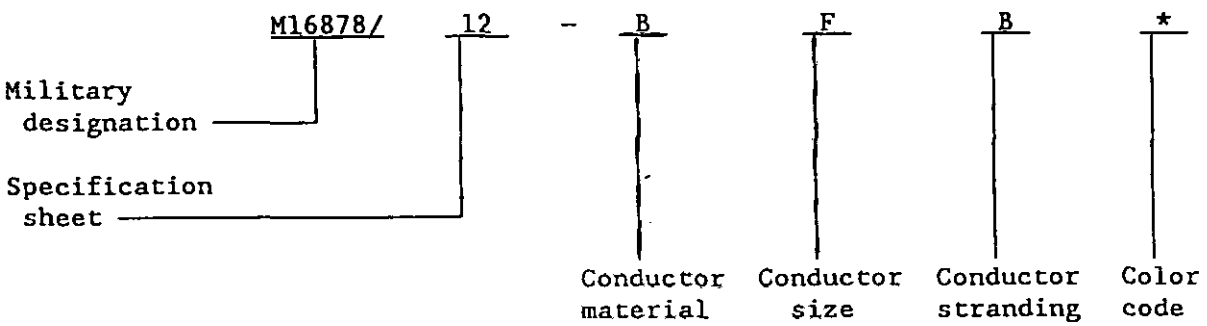
Wire size	Cold bend mandrel (maximum diameter in inches)
32 through 16	1
14, 12	2
10 through 6	3
4, 2	4.5
1, 0	6
00 through 0000	10

Concentricity: 70 percent (minimum).
Surface resistance: Not required.
Wrap back: Not required.
Heat resistance: Condition at 230°C.

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Flammability: Meets requirements of UL VW1 Flame Test.
Toxicity: Meets requirements of concentration levels specified in MIL-E-917.
Shrinkage: Required.
Heat aging: Not required.
Insulation tensile strength: 2000 pounds force per square inch (minimum).
Insulation elongation: 100 percent (minimum).
Marking and stripe durability: Required.
Fungus resistance: Not required.
Maximum dc resistance of finished wire: See table I of MIL-W-16878.
Identification of product: Required for wire sizes 22 AWG and larger.

PIN example (see MIL-W-16878):



Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6145-N330-11)

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